⌈Def. Let G be a group, and $H \leq G$ be a subgroup,
Given an element $g \in G$, define left-Coset of H as

$$gH \stackrel{\text{def}}{=} \{gh \mid h \in H\}.$$

⌈Lemma. Let G be a group, and $H \leq G$ be a subgroup. Then, for each $g_1, g_2 \in G$,

$g_1 H = g_2 H$ if and only if $g_1^{-1} g_2 \in H$.

Proof. Given $g_1, g_2 \in G$, suppose that $g_1 H = g_2 H$. Then, since $g_2 \in g_2 H = g_1 H$ $(1 \in H)$,

$g_2 = g_1 h$ for some $h \in H$, thus $g_1^{-1} g_2 \in H$.

Conversely, let $g_1, g_2 \in G$ be given s.t. $g_1^{-1} g_2 \in H$. That is, $g_1 h = g_2$ for some $h \in H$.
Given $g_1 h_1 \in g_1 H$, $g_1 h_1 = g_2 h^{-1} h_1 \in g_2 H$. Given $g_2 h_2 \in g_2 H$, $g_2 h_2 = g_1 h h_2 \in g_1 H$ $\square$

Q. In left Coset sence, $g_1 H = g_2 H \iff g_1 g_2^{-1} \in H$ hold?
A. No!, S_3, $H = \langle (1,2) \rangle$, $g_1 = (1\ 3)$ and $g_2 = (1\ 2\ 3)$

⌈Thm. Lagrange Thm
If H is a subgroup of a group G, then $|G|/|H| = [G:H]$
Prof. Given a subgroup $H \leq G$, define a relation on G as;

$$x \sim y \iff xH = yH$$

Then, it is an equivalence relation, so a Collection $\{gH \mid g \in G\}$ forms a partition.
Moreover, for each $g \in G$, a map

$$H \to gH \,;\, h \mapsto gh$$

is onto by definition and one to one by the Cancelation law, so $|H| = |gH|$.

Isomorphism Theorem.

1st Isomorphism theorem.

For any Homomorphism $\varphi : G \to G'$,

$$G/\mathrm{Ker}\,\varphi \cong \mathrm{Im}\,\varphi.$$

Proof. Since $\mathrm{Ker}\,\varphi \trianglelefteq G$, $G/\mathrm{Ker}\,\varphi$ is well-defined as a group. Define

$$\tilde{\varphi} : G/\mathrm{Ker}\,\varphi \to \mathrm{Im}\,\varphi : g\,\mathrm{Ker}\,\varphi \mapsto \varphi(g).$$

Then, $g_1 \mathrm{Ker}\,\varphi = g_2 \mathrm{Ker}\,\varphi \iff g_1 g_2^{-1} \in \mathrm{Ker}\,\varphi \iff \varphi(g_1) = \varphi(g_2)$

so $\tilde{\varphi}$ is well-defined and injective, and the surjective is clear. ∎

2nd Isomorphism theorem.

Let G be a group, and let A, B be subgroups of G satisfying $A \leq N_G(B)$.
Then, $AB \leq G$ and $B \trianglelefteq AB$, $A \cap B \trianglelefteq A$. Further,

$$AB/B \cong A/A\cap B$$

Proof. Since A normilizes B,

$$AB = \bigcup_{a\in A} aB = \bigcup_{a\in A} Ba = BA$$

and so given $a_1b_1, a_2b_2 \in AB$, $a_1b_1(a_2b_2)^{-1} = a_1b_1b_2^{-1}a_2^{-1} \in AB$, so $AB \leq G$.
$B \leq AB$ is trivial, and given $ab \in AB$ and $g \in B$, $abgb^{-1}a^{-1} \in B$, so $B \trianglelefteq AB$.
$A \cap B \trianglelefteq A$ is also trivial (These are given by the fact that A nomalizes B).
Now, define a map

$$\varphi : A \to AB/B : a \mapsto aB.$$

It is well-defined since $A \subseteq AB$ and it is Homomorphism because: given $a_1, a_2 \in A$,

$$\varphi(a_1a_2) = a_1a_2B = a_1B \cdot a_2B$$

and onto: given $abB \in AB/B$, take $a \in A$, then $\varphi(a) = aB = abB$.
And, the kernel is $\mathrm{Ker}\,\varphi = \{g \in A : gB = B\} = \{g \in A \mid g \in B\} = A \cap B$. Consequently,
the first isomorphism theorem gives $A/A\cap B \cong AB/B$. ∎

「3rd Isomorphism Theorem.

Let G be a group and $H, K \trianglelefteq G$ be normal subgroups with $H \leq K$.
Then, $K/H \trianglelefteq G/H$ and $[G/H]/[K/H] \cong G/K$.

proof. Since $K \trianglelefteq G$, $K/H \trianglelefteq G/H$ directly. Define a map

$$\varphi : G/H \to G/K : gH \mapsto gK.$$

It is well-defined, because: given $g_1, g_2 \in G$,

$$g_1H = g_2H \Leftrightarrow g_1g_2^{-1} \in H \underset{\substack{\Uparrow \\ H \leq K}}{\Rightarrow} g_1g_2^{-1} \in K \Leftrightarrow g_1K = g_2K = \varphi(g_1) = \varphi(g_2)$$

(so, injectiveness is given only $H = K$).

Surjectivity and Homomorphism are clear. Moreover,

$$\ker \varphi = \{gH \in G/H \mid gK = K\} = \{gH \in G/H \mid g \in K\} = K/H.$$

Finally, the 1st isomorphism theorem gives the result.